Aviation Analysis Business Presentation

A Data-Driven Approach to Selecting Low-Risk Aircraft

Faith Karimi Muthoni
DSF-FT12-HYBRID

Overview:

- Understanding aviation space, its risks and safety challenges
- Analysis of historical aviation accident data
- Key insights and business recommendations
- Next steps for improving aviation safety

Business Understanding

Problem Statement

- Our company is expanding into commercial and private aviation to diversify its portfolio.
- We lack insight into potential aircraft risks and need to determine the safest options for purchase.
- **Objective:** Identify the lowest-risk aircraft to help the new aviation division make informed purchasing decisions.

Why This Matters to the Business.

- Aviation accidents and incidents result in financial loss, reputational damage, and safety concerns.
- Understanding risk factors helps airlines, regulatory bodies, and insurers make data-driven decisions.
- **The goal:** Identify key risk factors and reduce aviation accidents.

Data Understanding

Data Used

Source: National Transportation Safety Board (NTSB)

Timeframe: 1962 - 2023

Key Variables:

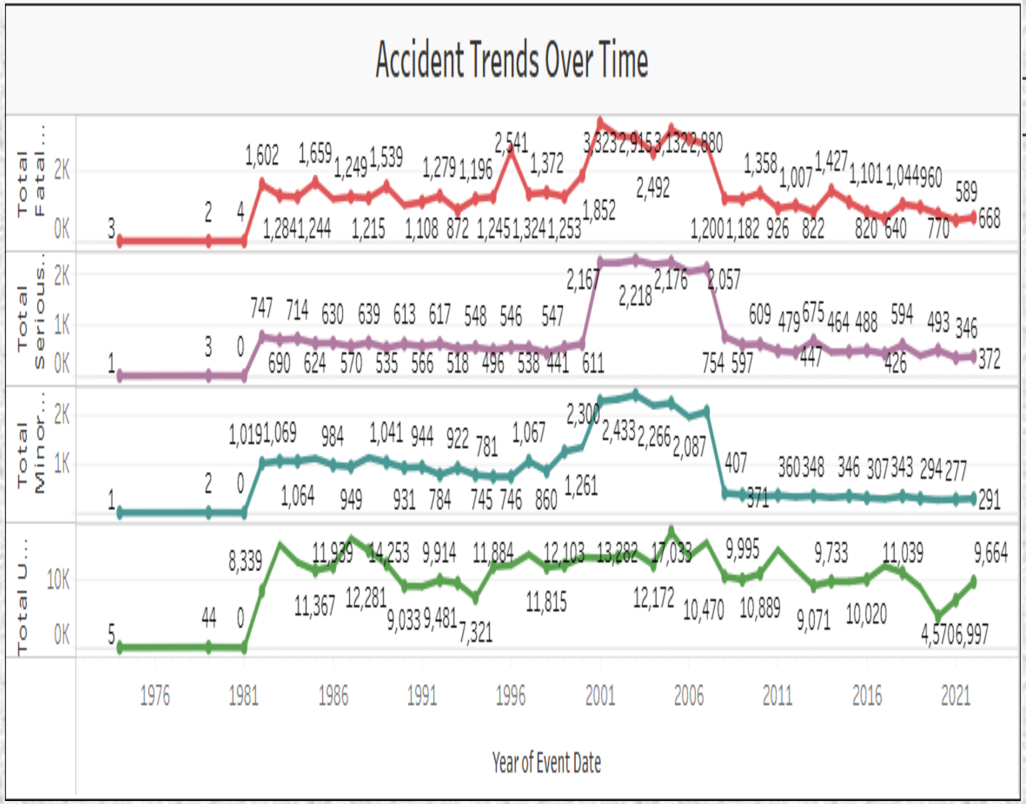
- Accident severity (fatal, non-fatal, incident)
- Engine type
- Manufacturer
- Purpose of flight

Data Limitations

- Some missing values in older records
- Possible reporting inconsistencies across years

Data Analysis

Key Findings & Visualizations



1. Accident Trends Over Time

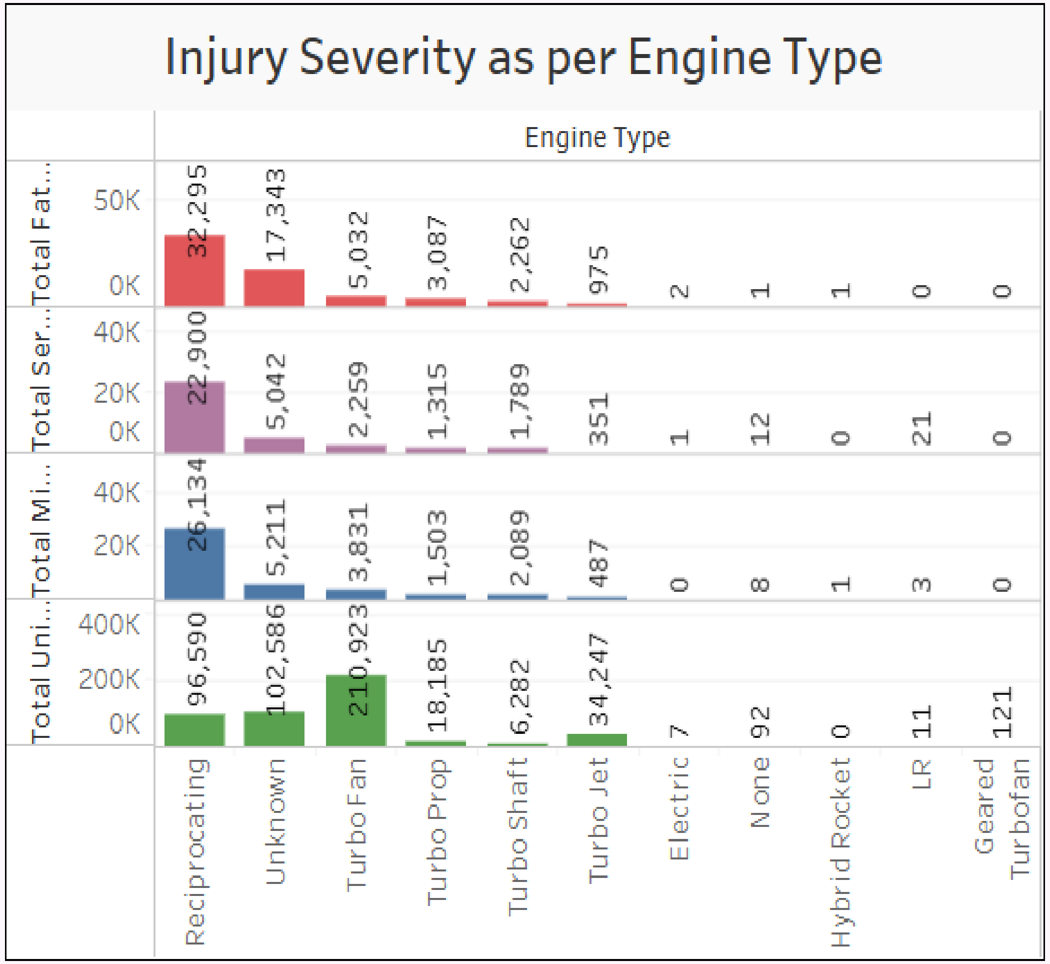
Insight:

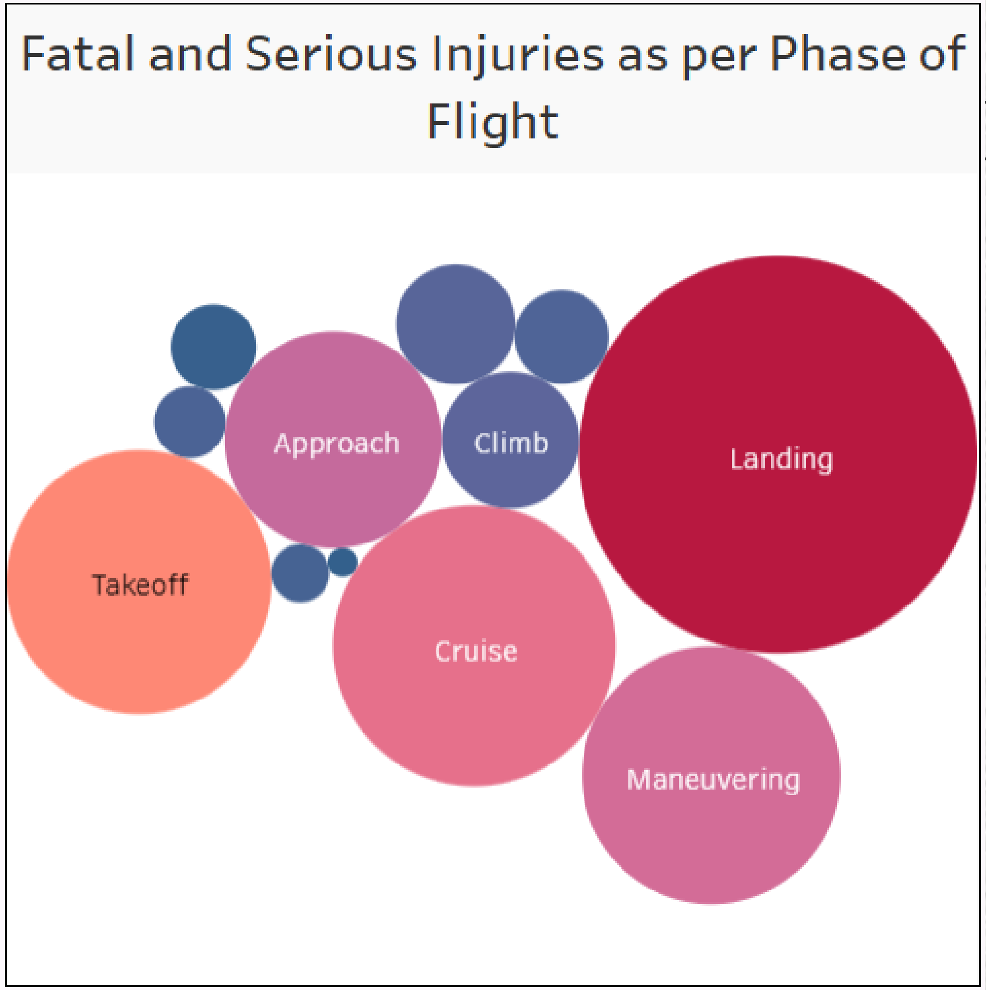
While the total number of aviation accidents has generally declined over the years, the number of fatal and serious injuries has fluctuated, indicating that while safety measures may have improved, severe incidents still occur.

2. Injury Severity as per Engine Type

Insight:

- Reciprocating engines have the highest number of total injuries, severe injuries, and fatalities, indicating higher risk.
- Turbojet and Turbofan engines report significantly fewer severe injuries and fatalities, highlighting their improved safety.
- Unknown and miscellaneous categories also show a notable number of incidents, likely due to incomplete reporting or diverse aircraft types.
- Electric, Hybrid Rocket, LR, and Geared Turbofan have recorded zero or very minimal injuries, indicating limited operational history or low adoption in commercial and general aviation.





3. Fatal & Serious Injuries as per Phase of Flight

Insight:

- The landing phase has the highest number of fatal and serious injuries, suggesting increased risk during touchdown procedures.
- Takeoff, cruise, and maneuvering phases also show significant injury numbers, indicating risks related to acceleration, in-flight operations, and low-altitude maneuvering.
- Climb and descent phases have fewer injuries compared to landing and takeoff, but unexpected turbulence or mechanical failures during these phases can still be hazardous.
- Taxi incidents, though less frequent in severity, suggest that ground operations are not entirely risk-free.

Business Actions Based on Findings

1. Invest in Safer Aircraft Models

- ✓ Prioritizing turbojet and turbofan aircraft for commercial operations enhances safety and minimizes financial liability.
- ✓ Reciprocating engine aircraft, often used in general aviation and smaller operations, may require stricter risk management and safety protocols.
- ✓ Investing in modern engine technology improves operational reliability, reducing long-term maintenance and insurance costs.
- ✓ Prioritize modern, proven aircraft models with safer engine configurations and advanced landing and takeoff assistance.

2. Enhance Predictive Maintenance Strategies

- ✓ Airlines and aviation authorities should focus on enhanced landing safety measures, such as improved runway conditions, automated landing assistance, and pilot training for difficult approaches.
- ✓ Takeoff and maneuvering risks highlight the importance of engine reliability, aircraft handling training, and in-flight safety monitoring.
- ✓ Safety enhancements in ground operations (e.g., better taxiway visibility, collision avoidance systems) can further reduce minor but preventable incidents.
- ✓ Understanding accident distribution across flight phases allows stakeholders to prioritize safety investments in the most high-risk areas.

3. Focus on Fleet Flexibility & Low Operational Risk

- ✓ For **private** aviation: Consider modern, fuel-efficient light jets and turboprops (low accident rates, cost-effective).
- ✓ For **commercial** operations: Mid-sized jets offer strong safety records and economic efficiency.
- ✓ **Pilot Training & Standard Operating Procedures (SOPs):** Invest in advanced flight training, emergency preparedness, and recurrent simulator training.
- ✓ **Predictive Maintenance Programs:** Use AI-driven monitoring systems to reduce mechanical failure risks and ensure aircraft remain airworthy.

Future Actions & Considerations

- Conduct **deeper analysis on specific aircraft manufacturers and models**.
- Collaborate with aviation safety organizations for **policy improvements**.
- Implement **automated risk assessment dashboards** for real-time monitoring.

THANK YOU

[GitHub Repository]

[Tableau]

faith.karimi@student.moringaschool.com